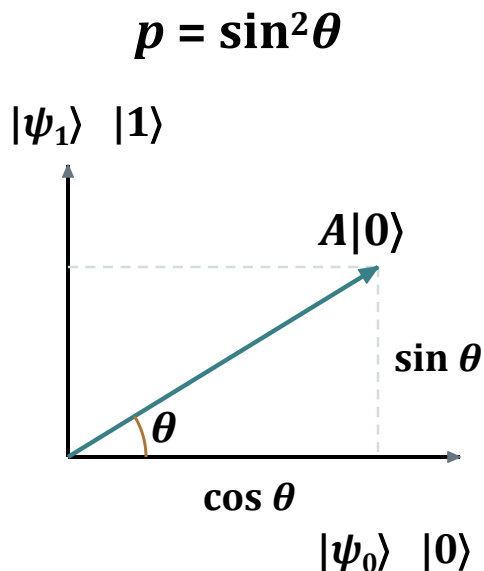
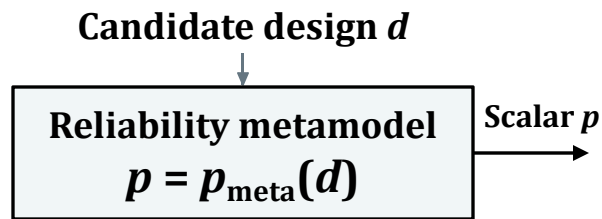


(a) Probability and amplitude

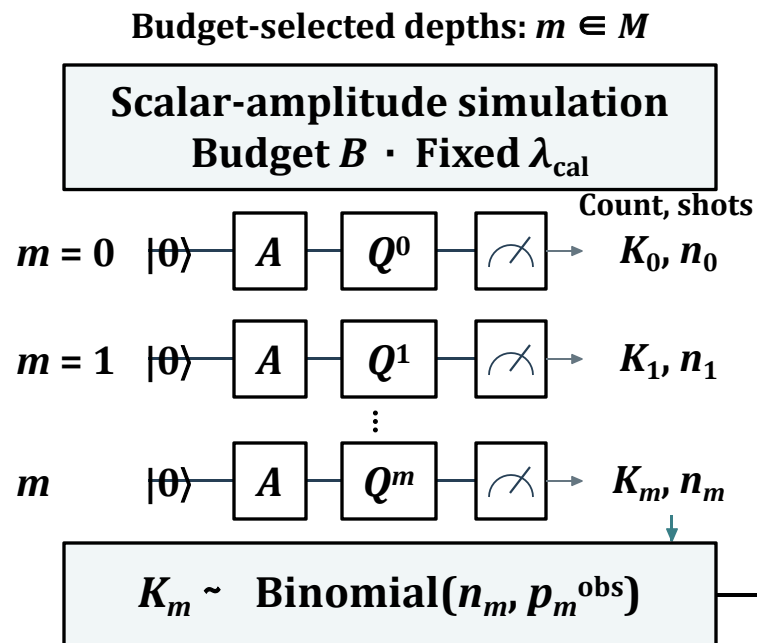


Amplitude geometry (schematic)

Ancilla $|1\rangle$: system exceedance

Estimator output to QAE-ABC

(b) Multi-depth count simulation



$$p_m^{\text{obs}} = 1/2 + v_m(p_m - 1/2)$$

$$p_m = \sin^2[(2m + 1)\theta]$$

$$v_m = \exp[-\lambda_{\text{cal}}(2m + 1)]$$

Depth-dependent visibility

(c) Likelihood and interval

Angle grid $\rightarrow$ Candidate peaks $\rightarrow$ Local refinement

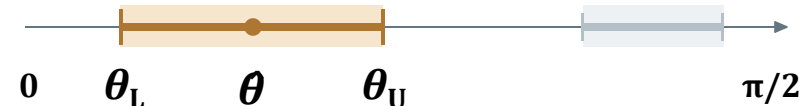
Calibrated maximum likelihood
 $\theta = \arg \max_{\theta} \ell(\theta; \lambda_{\text{cal}})$

Likelihood-ratio acceptance

$$2[\ell(\theta) - \ell(\theta)] \leq c_{\alpha}$$

c_{α} : χ^2 cutoff at $1 - \alpha_{\text{sig}}$ (1 d.f.)

Accepted components (schematic)



Retain the component containing θ

Selected angle interval: $[\theta_L, \theta_U]$

$$E(d; B, \alpha_{\text{sig}}) = [\hat{p}, L, U, b]$$

Point estimate

$$\hat{p} = \sin^2 \theta$$

Probability interval

$$[L, U] = [\sin^2 \theta_L, \sin^2 \theta_U]$$

Actual query use

$$b = \sum_{m \in M} n_m(2m + 1) \leq B$$